IMPERIAL

Supply Chain Game

Fourgotyourpackage

Freight Transport & Logistics – Group 4

24/02/2026

Kilian Schwarz
Nicolas Mancera
Yuya Ueda
Maria Sanchez
Sergi Marsol
Leen AlBinAli
Tala Matar
Ahad Hashmi

Agenda

1. Approach

The team applied a consistent four step methodology across both games

Games approach overview

1 Demand analysis

- Analyzed **historical demand** (trend, volatility, stockouts)
- Formed **forward assumptions** and **projected regional demand**

2 Strategy definition

- Defined **long-term** network design (factory and warehouse footprint and allocation)
- Set **short-term** operating levers (order quantity, ROP, shipping, fulfillment)

Deep-dive

3 Scenario testing

- Developed an **internal simulator** to model system dynamics
- **Stress-tested** baseline strategy under different demand, capacity and short-term **decision configurations**

4 Monitoring & updating

- Regularly reviewed **automated (bot) activities** to identify and resolve bugs
- Tracked **cash performance** against expectations and **logged learnings**

Decisions were structured into long-term and short-term
Splitting strategic design from rule-based execution

Decisions by round and type – Framework

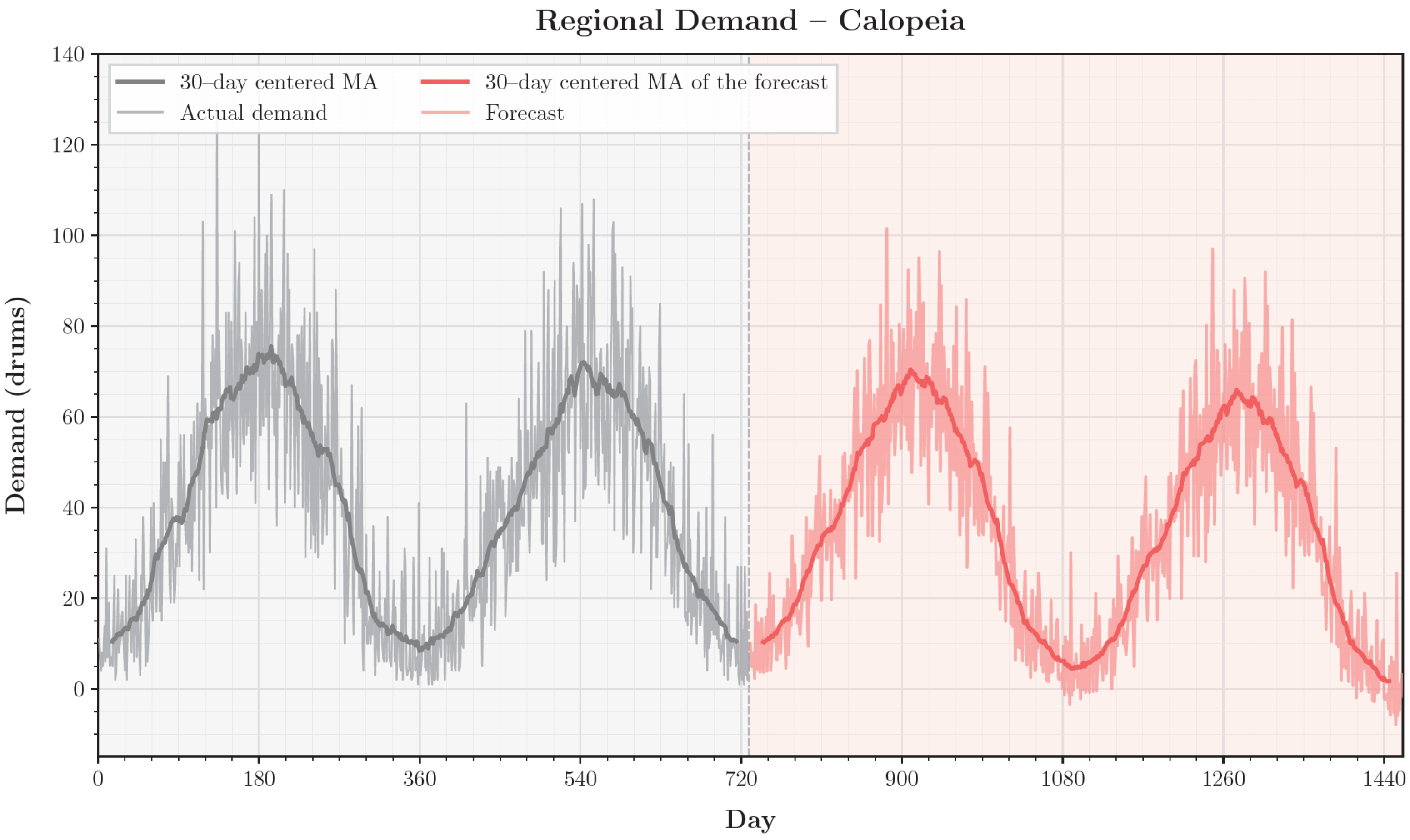
Deep-dive

Round / Type	Long-term	Short-term
Round 1	• Calopeia **factory capacity** • Calopeia **factory capacity activation**	• Calopeia **order quantity** • Calopeia **re-order point** • Calopeia **shipping method**
Round 2 *(Additional to round 1)*	• **Additional** factories • **Additional** factories **capacity** • **Factory-to-warehouse** allocation • Factories **capacity activation** • Factory **priority** • **Additional** warehouses • **Warehouse-to-region** allocation • Additional warehouses **activation**	• **Additional** factories **order quantity** • **Additional** factories **re-order point** • **Additional** factories **shipping method** • Warehouse **fulfillment policy**

2. Calopeia Round

Calopeia demand was forecasted through Holt-Winters additive method

Actual and forecasted demand



Key insights

Trend

- 365-day **seasonal cycle**, averaging **39.19 drums/day** with **peaks near 2x** the mean
- High percentage of **variability** is **predictable seasonality** rather than noise. Deseasonalised coefficient of variation[1]: $\boldsymbol{CV = \sigma_R / \bar{y} = 0.177}$ far lower than the **naive** $\boldsymbol{CV = 0.676}$

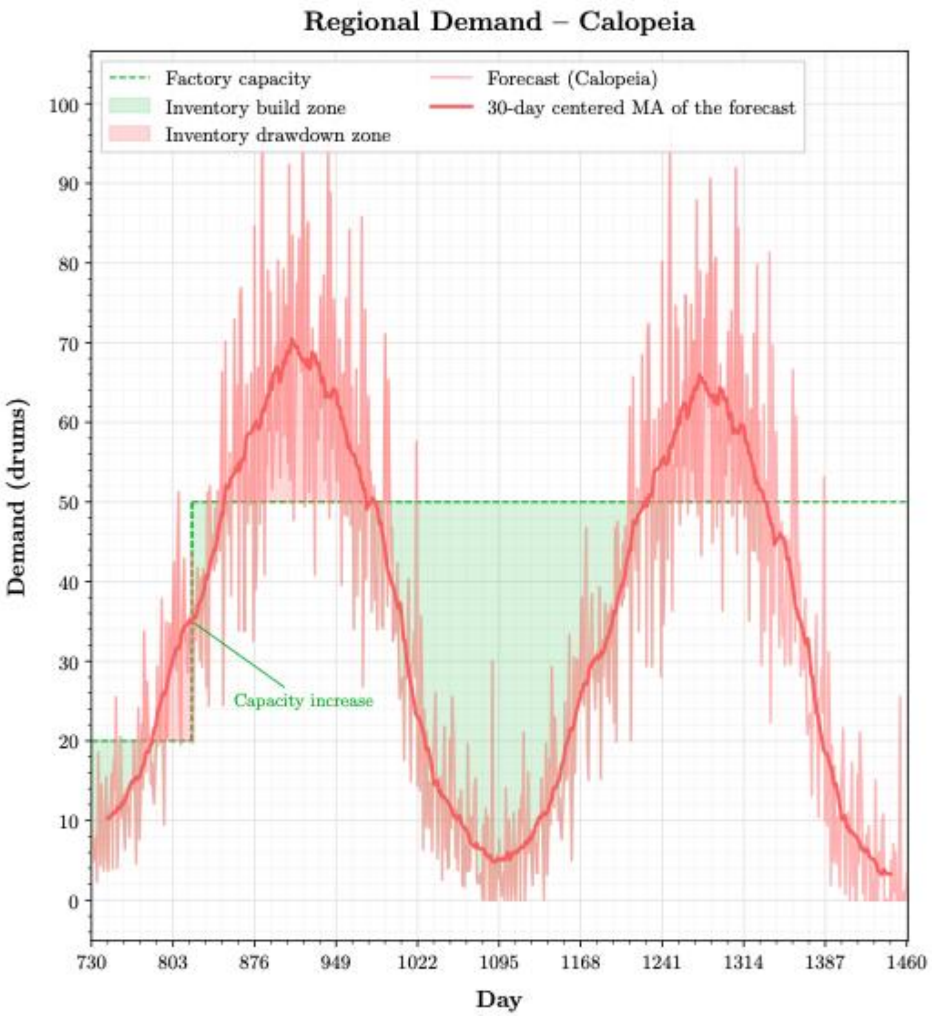
Projections

- Due strong seasonality, **Holt-Winters additive forecasting** was selected, capturing both seasonality and trend

Notes: 1) Computed from STL residuals

Demand analysis led to the development of a clear, executable strategy

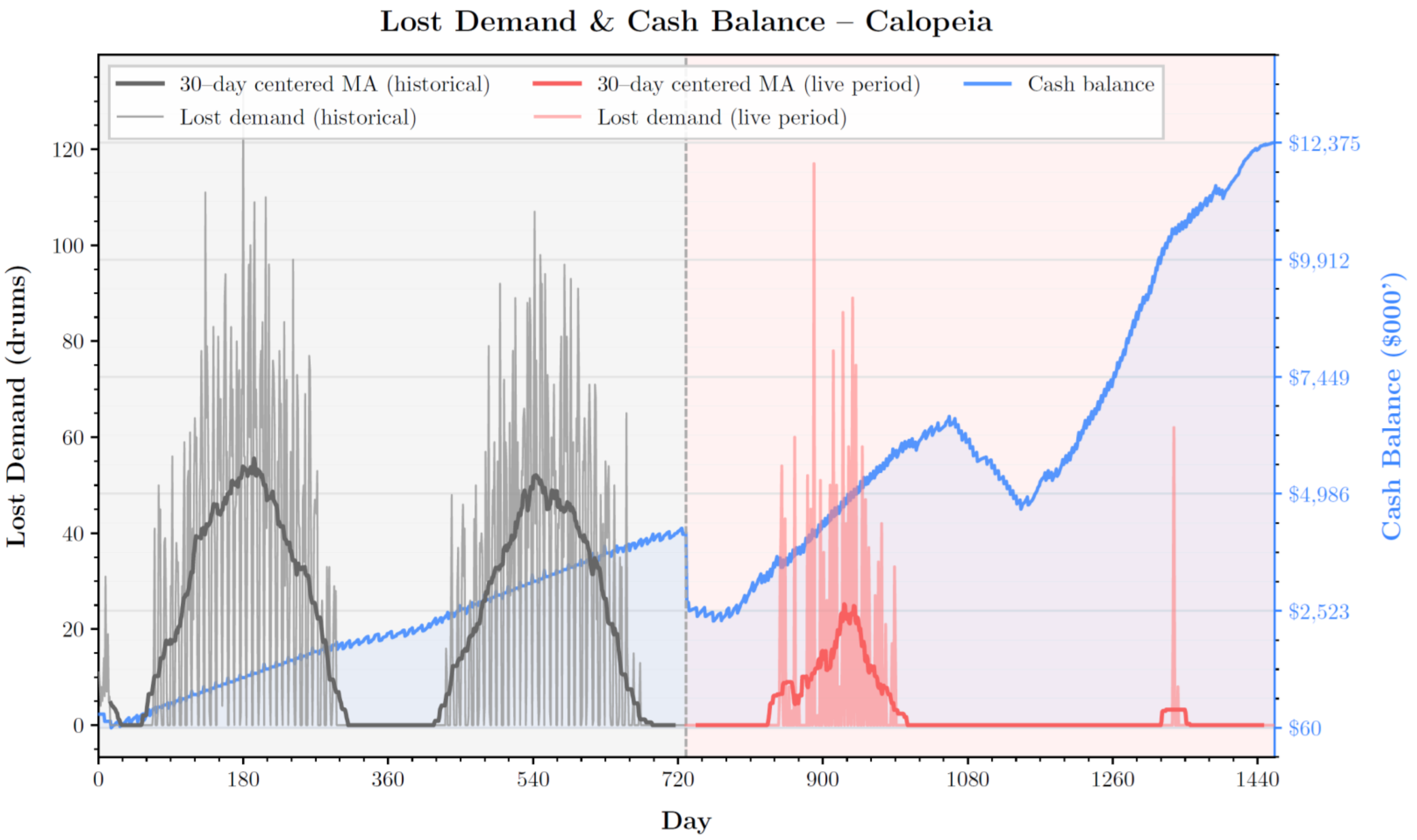
Key decisions, actions and rationale



Decision	Action and rationale
Capacity	**50 drums / day**, slightly above past demand mean
Activation	**Day 730** to **build inventory** before third peak
Re-order point (ROP)	**Dynamic ROP** across three modes: • **Drawdown:** 100% production, **deplete stock** to cover excess demand • **Build:** **Accumulate stock** beyond current demand when **future peaks** are expected • **Chase:** Match production to demand to maintain **stable inventory** levels
Order quantity	Use a **fixed truck batch** size of **200 drums**
Shipping method	Use **truck shipping** during normal play and switch to mail for **specific situations** (e.g., endgame)

Round 1 closed top-3, with cash well above the do-nothing output

Overview of ‘Calopeia Round’ Results



Key insights

Cash

- Cash grows steadily from **~$2.5M to $12.4M**
- **Final cash** trajectory is **steep**, confirming effective endgame liquidation

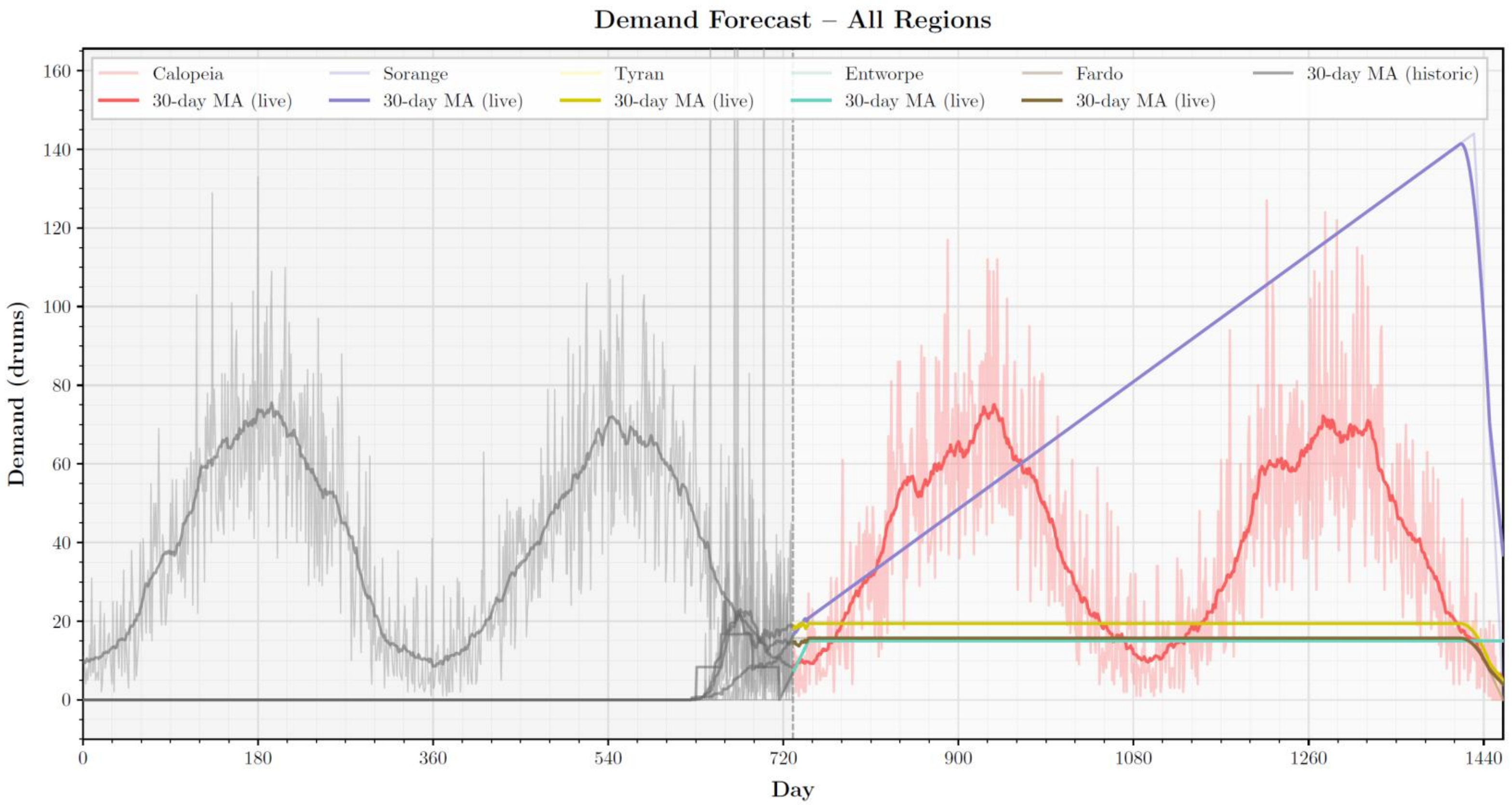
Lost demand

- **High demand** loss during days **850–1000** reflecting drawdown mode
- **Recovery** after day 1000 with **near-zero lost demand**
- **Unexpected late spike** (~day 1320) suggests a **brief stockout. Potential** for setting slightly **higher capacity**

3. Network Round

Tailored forecasting methods were needed for each target region

Actual and forecasted demand



Key insights

Trend

- Five regions with **distinct profiles**: Calopeia (seasonal), Sorange (linear growth), Tyran (stable), Fardo (stable), Entworpe (lumpy batch orders)

Projections

- **Calopeia and Sorange** forecast to **drive most network volume**, each requiring a dedicated method
- Tyran and Fardo projected stable, moving average applied
- **Entworpe** managed with fixed daily rate given lumpy orders
- **High noise** regions favour **simpler forecasting** methods

Network round required further splitting the decision framework

Decision making approach for network round

Updated decision framework

Decisions

Long-term

a) Network design

- **Additional** factories
- **Additional** warehouses

b) Network configuration

- Factory-to-warehouse **allocation**
- Factories **activation** timeline
- Factory **priority**
- Warehouse-to-region **allocation**
- Additional warehouses **activation** timeline

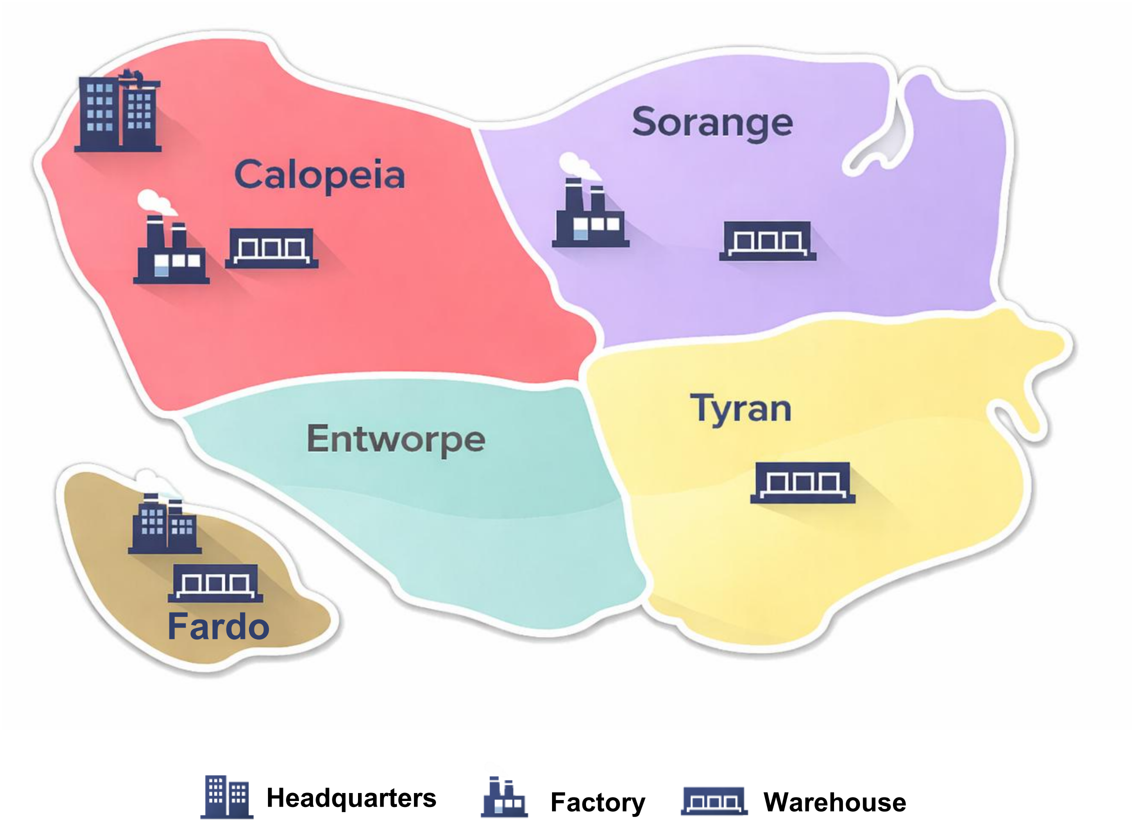
Short-term

- **Order quantity**
- **Re-order point**
- **Shipping** method
- Warehouse **fulfillment policy**

ROI analysis drove a three-system network to minimise shipping costs

a) Network design – Long-term

Established network and decisions



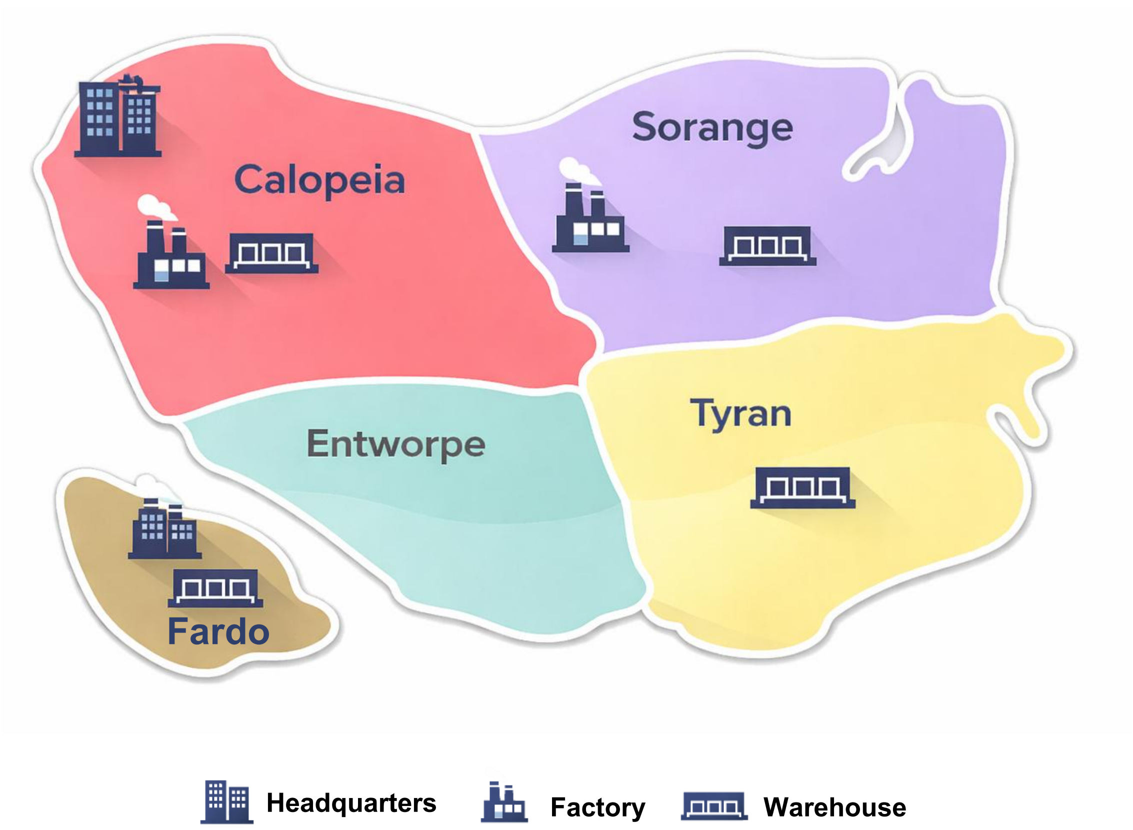
Three independent systems to **balance risk** and **minimise intercontinental shipping** costs (justified by ROI over 640 days)

1. **Calopeia system:** Manage well known Calopeia demand, Tyran WH and Entworpe via inter-region fulfilment
2. **Sorange system:** Focus on Sorange demand which grows linearly, and create some buffer to handle Calopeia's first peak
3. **Fardo system:** Focus on Fardo only

Capacity was sized to regional demand with coordinated activation plan

b) Network configuration – Long-term

Established network and decisions



Both **capacity and warehouse requirements** were **sized** and **timed** in alignment with **demand analysis insights**:

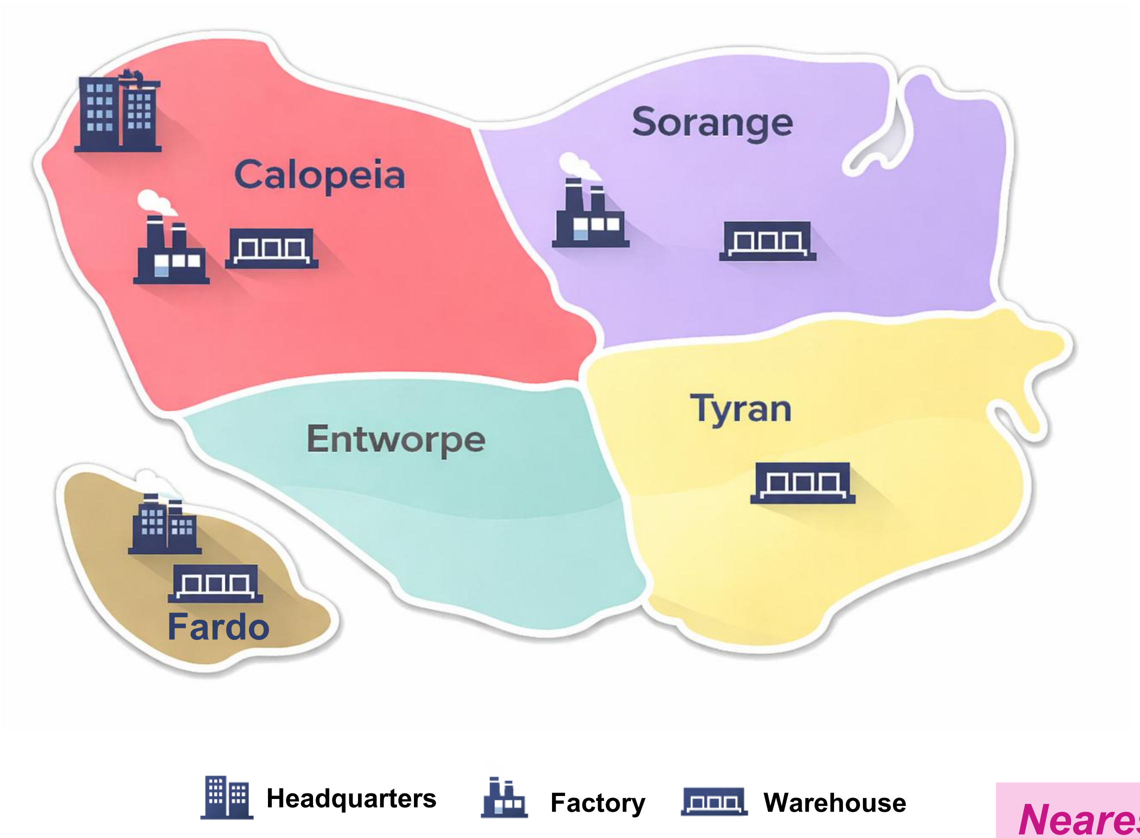
1 **Calopeia system**
- Factory produces **75 drums/day**
- New warehouse built in Tyran

2 **Sorange system**
- Factory produces **85 drums/day**
- New warehouse built in Sorange

3 **Fardo system** *(activated when cash $2.4mn)*
- Factory produces **20 drums/day**
- New warehouse built in Fardo

"Always On" production and variable batching maxed capital utilization
Short-term decisions

Established network and decisions



Short-term decisions applied **equally** to all systems:

Re-order point (ROP)
- Simplified in comparison to Calopeia Round
- ROP is always set 1,000 drums above current pipeline, to guarantee full capacity production

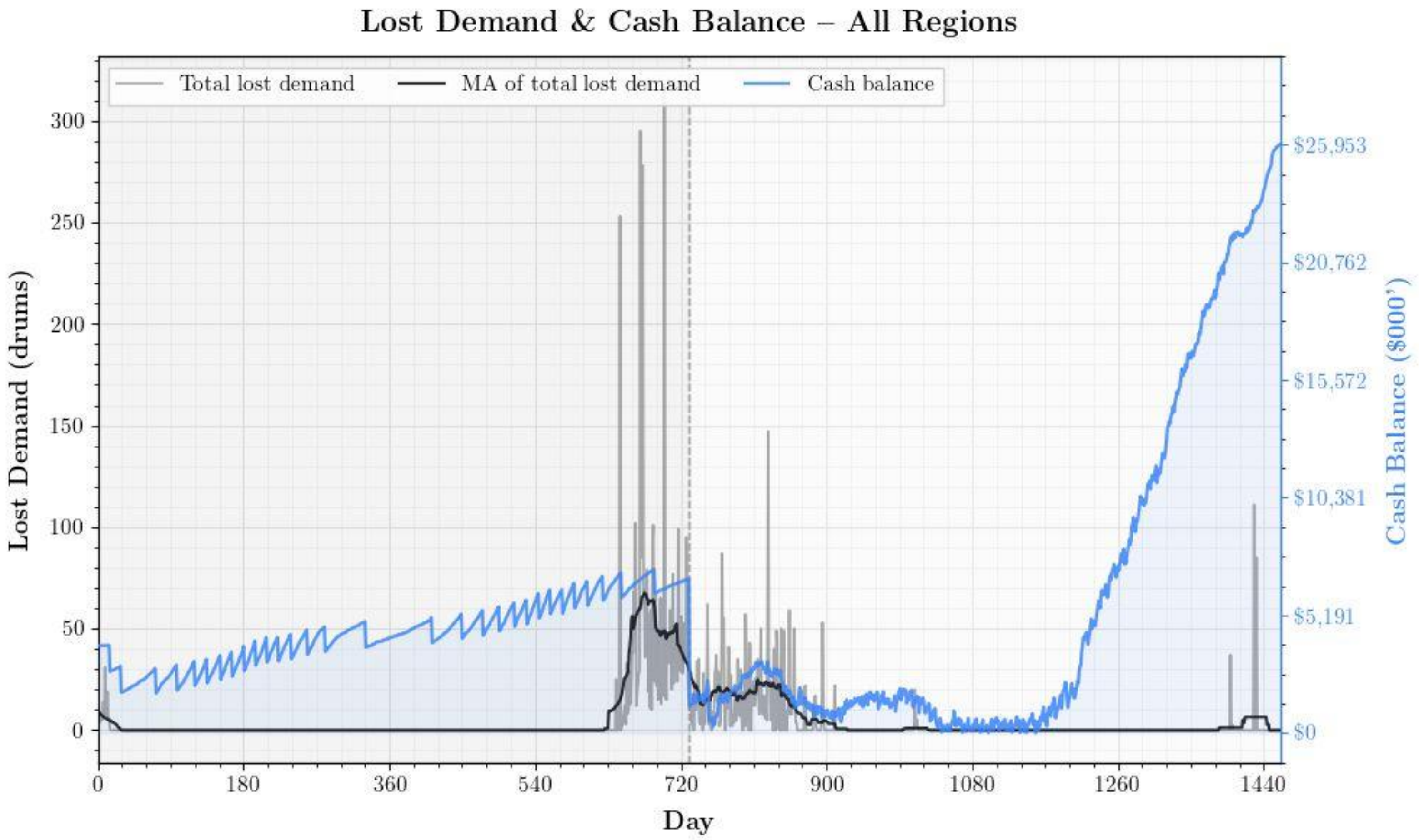
Order quantity
- **Variable batch** sizing **amortises** the $2,000 **fixed order cost.** Batches scale from 200 to 600 drums as inventory grows, reducing cost per drum

Shipping method
- Use truck shipping only. **Cross fulfilment** is **cheaper** and **more effective** than mail since it guarantees sales at only $50 more per barrel

Nearest fulfilment policy to avoid mainland-island cross-fulfilments

Strategy delivered a top finish, leveraging learnings from previous round

Overview of 'Network Round' Results



Key insights

Cash

- **Cash below starting ~$6.8M** for most of the game due to **capital expenditures** and **inventory** build-up
- Steep growth from day 1180 indicates **rapid liquidation** of **inventory**

Lost demand

- **Near-zero** from day **900** onward **validates** the "Always On" **overstocking strategy**
- **Late spike** (~day 1,420) reflects that a little more capacity buildup would have been advisable to improve item fill rate further

4. Conclusions

Conclusions

Iterating our way to a winning strategy

Strengths

- **Overstocking strategy** effectively **traded negligible holding costs** against **significant lost sales risk** ***(Network round item fill rate: 96.89%)***
- **Clear automation for short-term decisions** eliminated human error and enabled **consistent execution**
- **Simplifying operations** where **uncertainty** was **high** proved more **effective** than complex optimisation

Improvements

- **Surplus capacity** would have been prudent in both runs, instead of being overly parsimonious
- Utilizing **mail shipping** was ultimately unnecessary in Round 1. An ”Always on” strategy would have been superior
- **Batch sizing** could have been scaled dynamically from Round 1 to reduce fixed ordering costs

Q&A

Thank You

References & AI Statement

References

Escribano, J. and Angeloudis, P. (2026) *CIVE70108 Freight Transport and Logistics*. Imperial College London. Unpublished teaching material.

Merkle, M., Fusco, A., Jessen, P., Ravi, M. (2022) Capstone Project, Supply Chain Management Game: Final Report. Harvard Business School.

Generative AI statement

We acknowledge the use of generative AI to aid in specific tasks throughout this project. The Large Language Models (LLM) used were: Gemini Pro 3.0 (Google DeepMind, Google Gemini) to organize ideas during brainstorming sessions; Claude 4.5 Opus (Anthropic, Claude), to aid in troubleshooting and the programming of specific algorithms during the setup of the decision-making bot; Claude Sonnet 4.5 (Anthropic, Claude), to aid in the formatting of graphs; ChatGPT-5 (OpenAI, ChatGPT) to refine language in the present document, and to organize notes taken during team meetings. We confirm that these tools only played an auxiliary role in the development of this coursework, that all output was validated by the team, and that the content of this report was authored by members of the team.

Appendix A
Key formulas

Calopeia Round – Mathematical logic (1/3)
Economic & Inventory Logic

- Holding costs: $h = 90/365 \approx \$0.25$ per drum per day
- Profit per drum sold: $\pi = 1450 - \frac{2000}{200} - 900 - \frac{15{,}000}{200} - 150 = \$315$
- A drum can be held for 315/0.25 = 1,260 days

⇒ Given that holding costs are negligible relative to stockout costs, an upward inventory bias is economically justified.

Calopeia Round – Mathematical logic (2/3)

Inventory & Production Logic

- Safety stock: $SS = Z_{0.999} \cdot \sigma_d \cdot \sqrt{L_{eff}} \cdot 2$

  where $Z_{0.999} \approx 3.09$

  σ_d : The rolling 30-day demand standard deviation,

  L_{eff}: the effective lead time

⇒ Given the strong cost asymmetry between holding costs and stockout costs, the ×2 multiplier is economically justified.

Calopeia Round – Mathematical logic (3/3)

Inventory & Production Logic

The bot selects one of three modes each day.

Mode	Condition	Behaviour
DRAWDOWN	$\bar{d}_{14} > C$	Factory at 100%; deplete inventory to cover excess demand
BUILD	$\bar{d}_{14} \le C,\ D_{\text{fut}} > 0,\ I_t < I^*$	Accumulate stock for upcoming peaks
CHASE	otherwise	Steady state; production $\approx$ demand

$\bar{d}_{14}$: The mean forecasted demand over the next 14 days

I^* : The target inventory level ($= D_{\text{fut}} + SS$)

- ROP structure ensures early production under high service-level target.

$$\text{ROP} = \begin{cases} \displaystyle\sum_{i=1}^{L_{\text{eff}}} \hat{y}_{t+i} + SS & \text{Chase} \\ \max\left(D_{\text{fut}},\ \displaystyle\sum_{i=1}^{L_{\text{eff}}} \hat{y}_{t+i} + SS \right) & \text{Build} \\ 3000 & \text{Drawdown} \end{cases}$$

Network Round – Mathematical logic (1/3)

Network ROI Justification

Network design and ROI

Several factories and warehouse are built, each justified by ROI over 640 operating days:

Region	Decision	Capital Expenditure	Estimated ROI	Key Retional
Calopeia	Expand factory	-	Positive: core driver	Highest-demand continental hub; supplies Tyran & Entworpe; central production hub
Sorange	New factory + New warehouse	$600k Factory building and Warehouse building	~$424k	Growing demand (~150 dr/day peak); local production avoids cross-region shipping
Tyran	Warehouse only	$100k Warehouse building	~$101k	Local fulfilment saves $50/drum vs cross-region supply
Entworpe	No build	$0	Only $20.6k insufficient	Highly volatile demand (CV 4.43); dedicated WH not justified
Fardo	New factory + New warehouse	$1.6M Factory building and capacity expansion, Warehouse building	~$1.16M	Intercontinental shipping ($45k/truck) destroys margin if not localised

Network Round – Mathematical logic (2/3)

Economic Rationale (Why Overstock?)

Profit per drum sold: All cases [1) – 4)] are assuming 200-unit truck

1) Within-region fulfilment:

$$\pi = 1450 - \frac{2000}{200} - 900 - \frac{15{,}000}{200} - 150 = \$315$$

2) Cross-region factory to warehouse:

$$\pi = 1450 - \frac{2000}{200} - 900 - \frac{20{,}000}{200} - 150 = \$290$$

3) Warehouse cross-fulfilment on continent (Calopeia warehouse to Entworpe):

$$\pi = 1450 - \frac{2000}{200} - 900 - \frac{15{,}000}{200} - 200 = \$265$$

⇒ Given that holding costs are negligible relative to stockout costs, an upward inventory bias is economically justified.

Network Round – Mathematical logic (3/3)

Always-On Logic

- "Always On" reorder point

  Each factory uses a dynamic ROP that guarantees continuous production:

$$\mathrm{ROP}(t) = \underbrace{I_t + T_t}_{\text{pipeline}} + 1000$$

- Shutdown condition

  Production stops when inventory covers all remaining demand:

$$I_t + T_t \geq D_{\mathrm{rem}}(t) = \sum_{i=t+1}^{1450} \hat{d}^{\mathrm{adj}}(i)$$

⇒ Production continues at full capacity until remaining demand is covered, avoiding excess end-of-game inventory and protecting profitability.

Appendix B
Opportunities for Improvement

Calopeia Round – Detailed learnings

Opportunities for improvement

Non-exhaustive

Category	Decision	Issue	Improvement potential	Improvement details
Long-term	How much capacity for the factory?	50 BPD left demand unmet during the third seasonal peak	High	Slightly higher capacity (e.g. 55 BPD) would have prevented the brief stockout spike
	When to activate capacity?	Activation at day 730 left insufficient time to build inventory before peak	None	Limited by operations starting in Day 730
Short-term	What should be the order quantity?	Fixed 200-drum batches potentially inefficient when inventory was high during build phase	Medium	Variable batch sizing (such as that adopted in Round 2) could have reduced fixed ordering costs
	What should be the re-order point (ROP)?	Drawdown ROP triggered too aggressively around days 850–1,000, causing excess lost demand	Medium	Higher safety stock multiplier or earlier switch to Build mode could have reduced stockouts
	What should be the shipping method?	Mail switch timing in endgame not perfectly calibrated meaning small leftover inventory remained	Low	Tighter endgame demand forecast could have achieved better results (already strong end strategy)

Network Round – Detailed learnings

Opportunities for improvement

Non-exhaustive

Category	Decision	Issue	Improvement potential	Improvement details
Network Design (Long-term)	Additional warehouses	Entworpe WH not built	Low	Dedicated buffer WH could have absorbed lumpy demand spikes at days 550–720
	Additional factories	Fardo delayed by cash constraint	Low	Slightly earlier activation could have prevented lost demand during days 600–900
Network Configuration (Long-term)	Factories capacity	Fardo undersized	Low	Higher capacity could have been built to tackle Fardo a bit more at the beginning
	Factory priority	Tyran floor too low (400 drums)	Medium	Higher floor for Tyran could have prevented stockout spikes at days ~730 and ~1,260
	Factories activation timeline	Sorange ramp-up mistimed	Medium	Better synchronisation with demand growth curve could have prevented day ~700 spike of Sorange
Short-term	Re-order point	Calopeia ROP insufficient ahead of seasonal peak	Low	Higher buffer target could have prevented day drawdown spike for Calopeia
	Shipping method	No mail switching across regions	Low	Mail fallback could have plugged late stockouts in Tyran (~day 1,260) and Fardo (~day 1,050)

IMPERIAL